IntelliCredit: AI Model & Architecture Summary

Brief overview of the workflow, model selection, and core differentiators

1. End-to-End Workflow

The system operates through an automated 8-step pipeline, from the moment a Credit Officer uploads documents until the final Credit Appraisal Memo (CAM) is generated.

Phase	Module Components	Description
1. Ingestion	PDF Parser, Regex Engine	Extracts structured tables and unstructured text from Annual Reports, GST, and Bank Statements. Implements dual-track extraction (Regex + AI) for high reliability.
2. Financial AI	Claude 3.5 Sonnet	Structured JSON extraction identifies 30+ financial metrics. Crucially handles bracket negatives, Lakhs/Crores normalizations, and multi-column year layouts.
3. Cross-Verify	GST vs Bank Logic	Detects revenue inflation, matching credit-debit circular trading (> Rs 1L), and GSTR-2A vs 3B ITC mismatches to flag potential fraud.
4. Research	Web Crawler	Automated secondary research via DuckDuckGo/Google covering news, promoter history, MCA filings, and legal disputes (NCLT context).
5. NLP Engine	Custom Sentiment Lexicon	Extracts 6 categories of risk signals and runs sentiment analysis using a tailored 100+ Indian credit keyword lexicon.
6. ML & Scoring	XGBoost + Five Cs	Calculates 15 credit ratios. Runs an XGBoost risk classifier. Feeds into a weighted Five Cs of Credit model (Character, Capacity, Capital, Collateral, Conditions).
7. Decision	Credit Rules Engine	Outputs loan recommendations (Approve/Refer/Reject) based on final score, adding specific risk conditionality and rate guidance.
8. CAM Report	ReportLab, Python-Docx	Compiles all findings into a professional, heavily formatted PDF/Word Credit Appraisal Memo with automatic SHAP explainability visualizations.

2. Model Selection: What We Chose & Why

We architected IntelliCredit using a best-of-breed approach, opting for specialized models for specific tasks rather than relying on a single generic LLM.

Model Choice	Primary Function	Why we chose it over the alternatives
Claude 3.5 Sonnet	Financial Extraction from PDFs	**Why chosen:** Superior performance on complex, messy Indian spatial layouts and tables compared to GPT-4o. Extremely reliable at outputting perfectly enforced JSON schemas. **Alternative rejected:** Open-source LLMs (LLaMA) lacked the accuracy for zero-shot tabular spatial reasoning. Tesseract alone fails on native PDFs.
GPT-4 Omni	Narrative & CAM Generation	**Why chosen:** Best-in-class generative reasoning and narrative formulation. Used specifically to synthesize hard facts into the professional, coherent executive summaries required by credit managers.
XGBoost Classifier	Binary Credit Risk Prediction (Default/No-Default)	**Why chosen:** The industry gold standard for handling tabular financial data with missing values. Fast, interpretable, and doesn't overfit like deep neural networks. Built-in support for SHAP. **Alternative rejected:** Deep Learning (overkill, black-box) and Random Forest (XGBoost consistently outperforms).
Custom NLP Lexicon	Sentiment & Risk Signals	**Why chosen:** We built a ~100 keyword custom dictionary specific to the Indian lending context (NCLT, SFIO, CBI, wilful defaults). It is computationally deterministic and infinitely faster than running LLM sentiment on thousands of scraped news URLs. **Alternative rejected:** FinBERT (Heavy inference overhead, lacks specific Indian regulatory context).

3. Why It Is Important & Different

The Indian mid-market corporate lending sector operates on massive, disjointed datasets. IntelliCredit differentiates itself strictly through **Contextual Intelligence** and **Fraud Prevention**.

Key Differentiator	How IntelliCredit Operates
Dual-Track Extraction	Traditional models rely solely on either OCR or LLM prompts. We use both. A deterministic Regex-engine calculates algebraic truths (e.g., Revenue Growth) while the LLM parses the nuance (e.g., extracting specific 'attributable' Net Worth).
Real-time Fraud Checks	Instead of post-facto audits, the system automatically checks for Circular Trading (>1L matching transactions) and GSTR-2A vs 3B Mismatches at the start of the appraisal process.
Indian Market Nuance	Automatically normalizes numbers between Lakhs, Crores, and Millions found in the same document. Properly handles uniquely Indian structures (PPOP for NBFCs, RBI/SEBI regulatory web scraping).
Complete Explainability	Every AI decision is fully transparent. The ML model outputs a SHAP waterfall chart explaining feature importance, and the Five Cs model clearly traces every point gained or penalized back to its specific factor.

Impact on the Business

- **Radical Efficiency:** Turns a 2-to-3 week manual appraisal drag into a < 5 minute automated flow.
- **Reduced Human Error:** Replaces subjective assessment with a highly disciplined, 100% standardized checklist (e.g., mathematically confirming that PAT Margin < 150%, or GNPA > NNPA).
- **Preventative Defaulting:** By catching early warning signals via web scraping before manual underwriting begins, the NPL (Non-Performing Loan) ratio is systematically reduced.